

HAUS GABLES

JENNIFER BONNER - 2018 - ATLANTA, GEORGIA
HUMID CUBTROPICAL CLIMATE

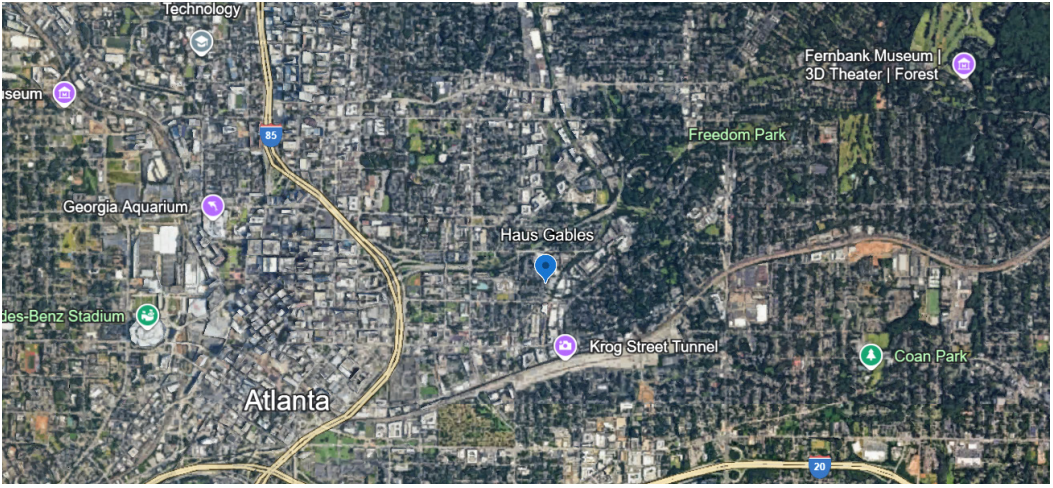
DESIGN PRIORITIES & STRATEGIES

Haus Gables is located in Atlanta, Georgia, a city characterized by a humid subtropical climate (Cfa) with hot, humid summers and mild winters. According to the Köppen climate classification, 'C' represents a warm temperate climate, 'f' indicates full humidity (no dry season), and 'a' signifies hot summers.

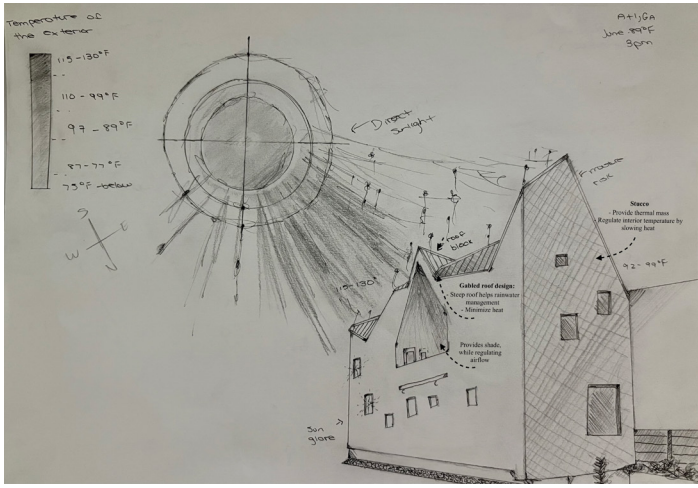
Summer temperatures may peak at 89°F, whereas winter lows typically hover around 45°F. The area receives considerable solar radiation throughout the summer, heightening the need for cooling, while winter calls for efficient heat retention methods. Elevated humidity levels lead to moisture accumulation, affecting material longevity and indoor comfort. Wind patterns change with the seasons, as predominant west winds during summer deliver warm, moist air, while northeast winds in winter bring in cooler, drier weather. While wind speeds are usually minimal, periodic storms elevate airflow, affecting passive ventilation methods.

Haus Gables incorporates climate-responsive designs into its material options, orientation, and massing. The structure is made of cross-laminated wood (CLT), with thermal mass qualities that assist inside temperatures. Its steep gabled roof effectively drains rainfall while minimizing solar heat gain. The structure's northeast-facing direction maximizes passive cooling, while strategically positioned windows improve natural ventilation and sunshine penetration. Additionally, insulating materials like Kingspan Kooltherm reduce heat loss and moisture retention.

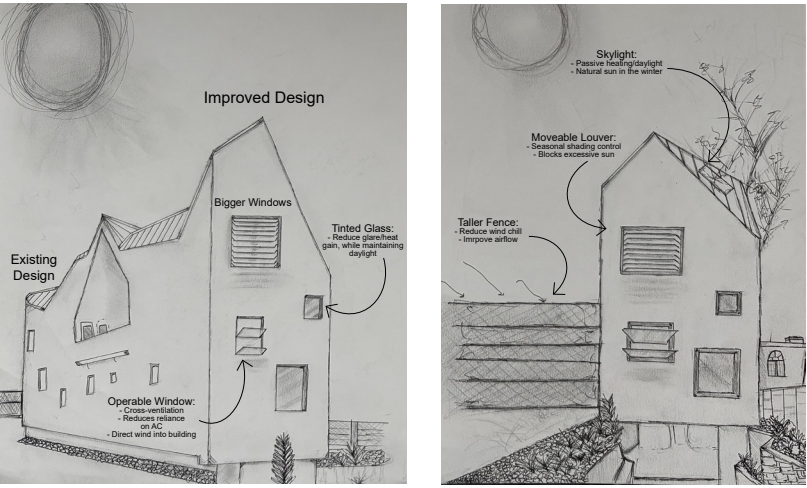
Haus Gables efficiently uses thermal mass and daylight strategies, with its elongated shape and open floor design encouraging air circulation for natural cooling. The CLT panels stabilize temperatures, while Kingspan Kooltherm insulation enhances energy efficiency and moisture management. The ridged roof promotes passive ventilation, and adding movable louvers, operable windows, and exterior shading, such as reflecting coatings, would improve circulation while reducing heat input. While the location lacks vegetation, adding passive cooling technologies and wind protection will enhance the building's performance in Atlanta's humid climate, ensuring energy efficiency and comfort year-round.



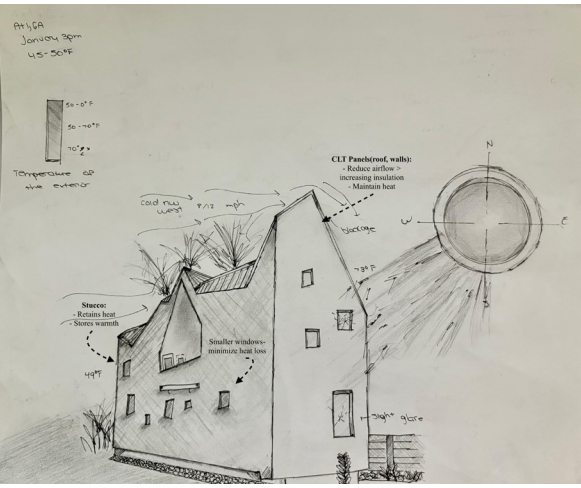
ZOOMED OUT MAP



SUN EXPOSURE, SHADING, AND SURFACES TEMPERATURES IN JUNE AT 3PM



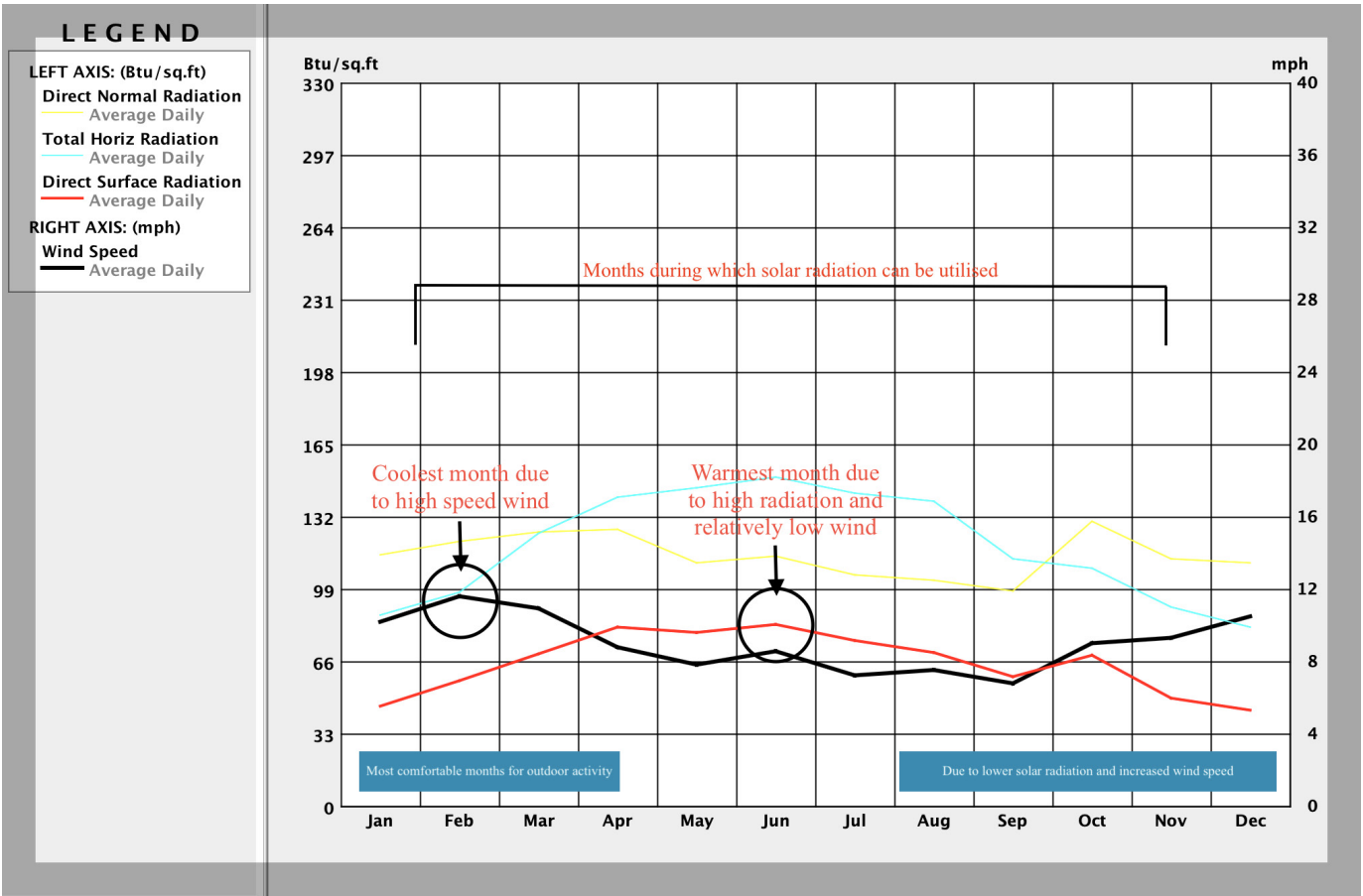
EXISTING & IMPROVED DESIGN WITH SHADING, VENTILATION, AND COOLING STRATEGIES



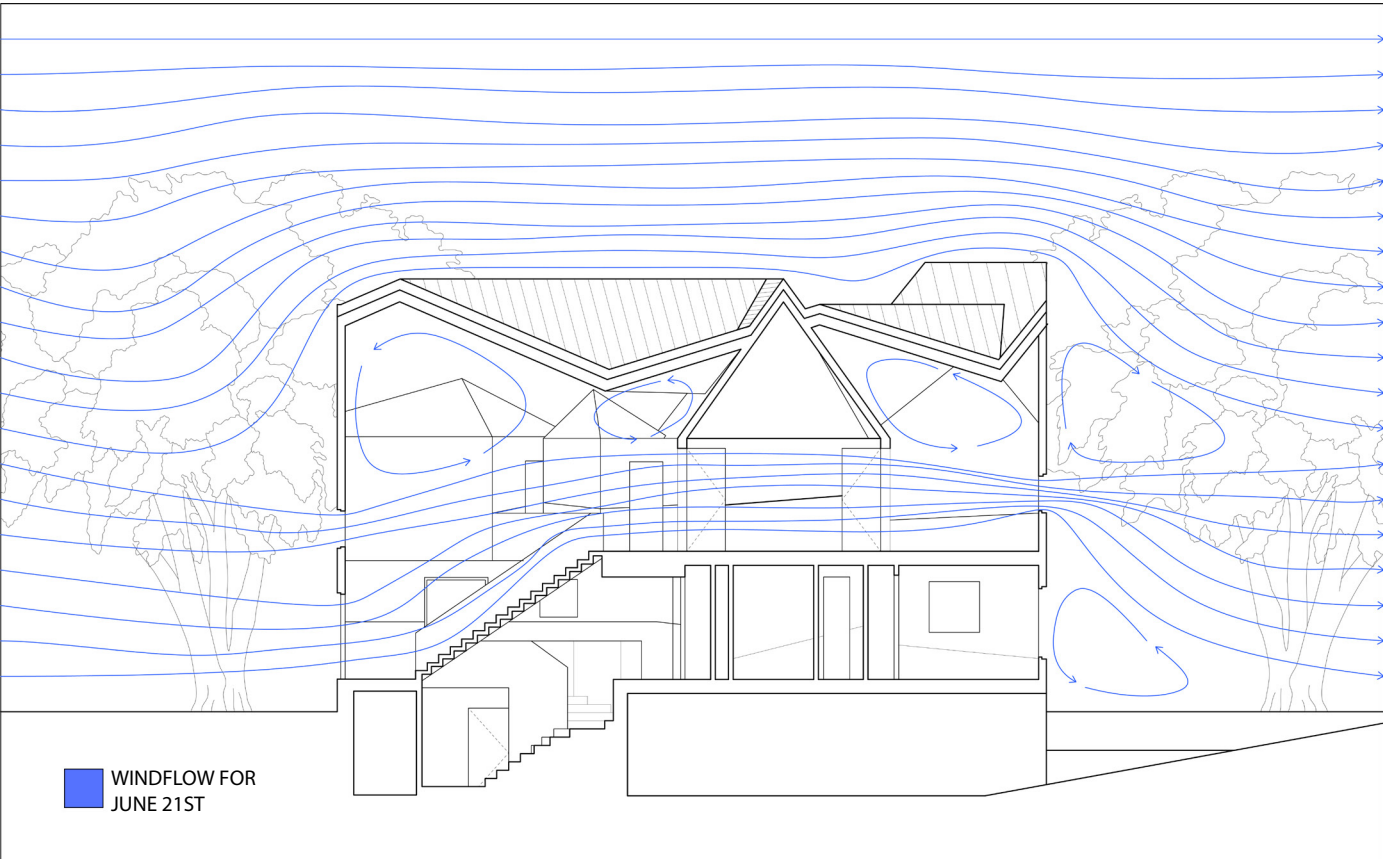
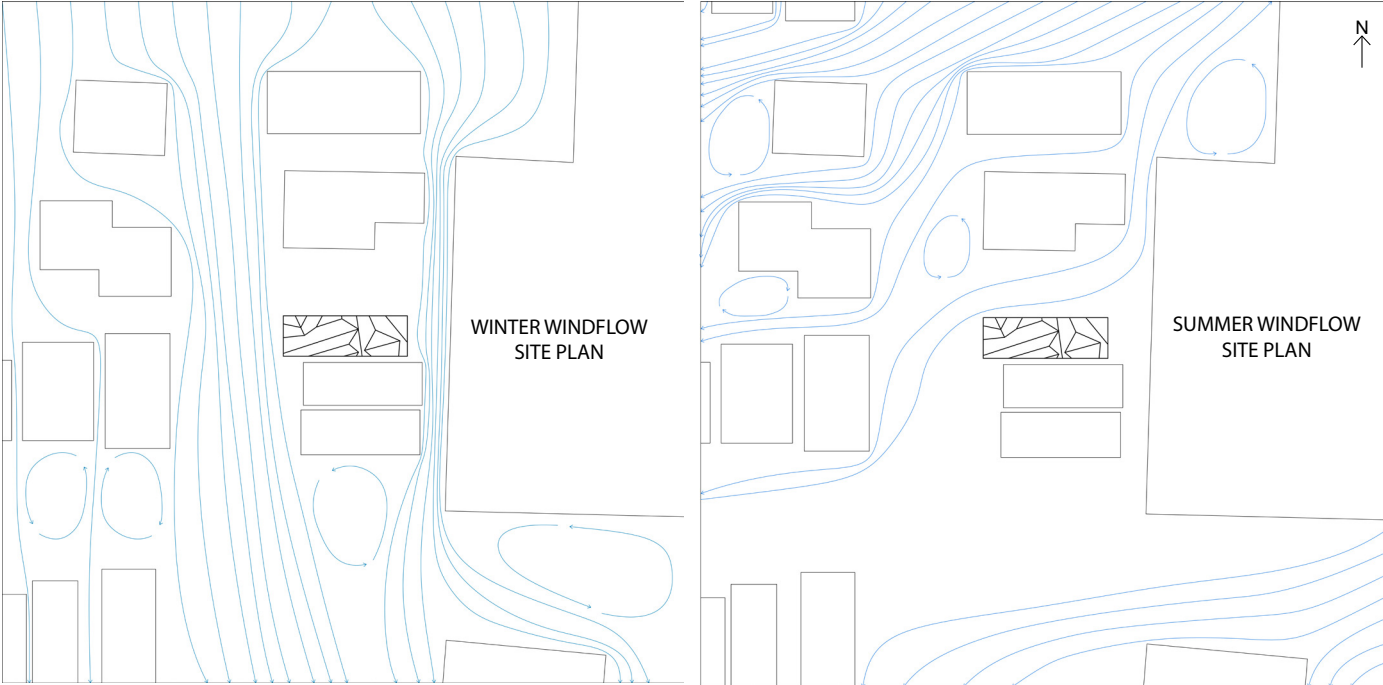
SUN POSITION, SHADOWS, AND HEAT RETENTION IN JANUARY 3PM



ZOOMED IN MAP



TEMPERATURE AND SOLAR CHARACTERISTICS



KEY IMAGES OF CASE STUDY BUILDING:



- REFERENCES:
- https://www.archdaily.com/916457/haus-gables-mall/5cd0595e284dd17d310000c7-haus-gables-mall-detail-section?next_project=no
 - <https://jenniferbonner.com/01-Haus-Gables>
 - <https://www.usclimatedata.com/climate/atlanta/georgia/united-states/usga1761>
 - https://rswweather.com/climate/Georgia/Atlanta/#google_vignette
 - <https://www.solarenergylocal.com/states/georgia/atlanta/>
 - <https://www.olinger.io/haus-gables>
 - <https://www.kingspan.com/us/en/case-studies/haus-gables/>



SUN SHADOW SITE PLANS